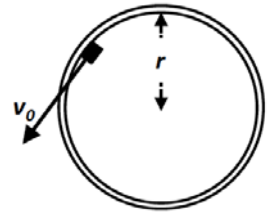


## University Physics Quiz 2

Name: \_\_\_\_\_ SID: \_\_\_\_\_ School/Department: \_\_\_\_\_

1. A block of mass  $m$  is sliding on a frictionless horizontal table. It is constrained inside a ring which is fixed on the table and of radius  $r$ . At  $t = 0$ , the block is moving along the inside of the ring with tangential velocity  $v_0$ , as shown in the accompanying figure. The coefficient of kinetic friction between the inner surface of the ring and the block is  $\mu$ .



(a) Find the speed of block at any later time  $t$ .

(b) Find the position of the block at any later time  $t$ .

Solution: (a) The equation of motion is

$$-e_r: \quad m \frac{v^2}{r} = N, \quad (1)$$

$$e_\theta: \quad m \frac{dv}{dt} = -\mu N. \quad (2)$$

Substituting Eq.(1) into (2), we have

$$m \frac{dv}{dt} = -\mu m \frac{v^2}{r} \quad \text{or} \quad -\frac{dv}{v^2} = \frac{\mu}{r} dt.$$

Integrating both sides of above equation, we obtain

$$\frac{1}{v} - \frac{1}{v_0} = \frac{\mu}{r} t \quad \text{or} \quad v = \frac{v_0}{\left(1 + \frac{\mu v_0 t}{r}\right)}.$$

(2) Using  $r \dot{\theta} = v$ , we have  $\frac{d\theta}{dt} = \frac{v}{r} = \frac{v_0/r}{\left(1 + \frac{\mu v_0 t}{r}\right)}.$

Then

$$\int_{\theta_0}^{\theta} d\theta = \int_0^t \frac{v_0/r}{\left(1 + \frac{\mu v_0 t}{r}\right)} dt = \frac{1}{\mu} \ln\left(1 + \frac{\mu v_0 t}{r}\right) \quad \text{or} \quad \theta = \frac{1}{\mu} \ln\left(1 + \frac{\mu v_0 t}{r}\right) + \theta_0.$$

2. A projectile of mass  $m$  is launched from the surface of the earth at an angle  $\alpha$  from the vertical. The initial speed is  $v_0 = \sqrt{\frac{GM_e}{R_e}}$  where  $M_e$  and  $R_e$  are the mass and radius of

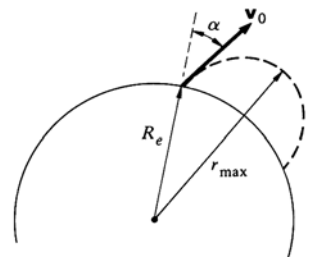
the earth respectively. How high does the projectile rise? (The resistance of the air and the rotation of the earth can be neglected.)

Solution: The conservation of the angular momentum reads

$$mR_e v_0 \sin \alpha = mvr_{\max}. \quad (1)$$

The conservation of mechanical energy reads

$$\frac{1}{2}mv_0^2 - \frac{GM_e m}{R_e} = \frac{1}{2}mv^2 - \frac{GM_e m}{r_{\max}}. \quad (2)$$



From eq.(1), we have  $v = \frac{v_0 R_e \sin \alpha}{r_{\max}}$ . Substituting  $v = \frac{v_0 R_e \sin \alpha}{r_{\max}}$  into eq.(2) we obtain

$$\frac{1}{2} v_0^2 - \frac{GM_e}{R_e} = \frac{1}{2} v_0^2 \frac{R_e^2}{r_{\max}^2} \sin^2 \alpha - \frac{GM_e}{r_{\max}}.$$

Noting that  $v_0^2 = \frac{GM_e}{R_e}$ , we obtain

$$-\frac{1}{2} \frac{GM_e}{R_e} = \frac{1}{2} \frac{GM_e}{R_e} \frac{R_e^2}{r_{\max}^2} \sin^2 \alpha - \frac{GM_e}{r_{\max}} \quad \text{or} \quad \sin^2 \alpha \left( \frac{R_e}{r_{\max}} \right)^2 - 2 \frac{R_e}{r_{\max}} + 1 = 0.$$

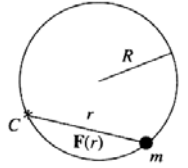
Then,

$$\frac{R_e}{r_{\max}} = \frac{2 \pm \sqrt{4 - 4 \sin^2 \alpha}}{2 \sin^2 \alpha} = \frac{1 \pm \cos \alpha}{\sin^2 \alpha}$$

Therefore, (noting that  $r_{\max} > R_e$ )

$$\frac{r_{\max}}{R_e} = \frac{\sin^2 \alpha}{1 - \cos \alpha} \quad \text{or} \quad r_{\max} = \frac{\sin^2 \alpha}{1 - \cos \alpha} R_e.$$

3. A particle of mass  $m$  moves in a circular orbit of radius  $R$  under the influence of a central force  $F(r)$ . The center of the central force lies at a point on the circle, as shown in the accompanying figure. What is the force law of the central force? (Hint: the equation of the orbit can be written as  $r = 2R \cos \theta$ .)



Solution: The equation of motion of the particle can be written in the component form as

$$\begin{aligned} m(\ddot{r} - r\dot{\theta}^2) &= F(r), \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) &= 0. \end{aligned}$$

The integrating of the second equation will lead to

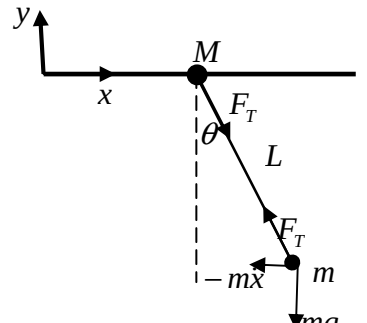
$$mr^2\dot{\theta} = L = \text{Const.}$$

Thus

$$\begin{aligned} F(r) &= m(\ddot{r} - r\dot{\theta}^2) = -m \left( \frac{d^2 r}{d\theta^2} \dot{\theta}^2 + \frac{dr}{d\theta} \ddot{\theta} - r\dot{\theta}^2 \right) = -m \left( \frac{d^2 r}{d\theta^2} \dot{\theta}^2 - \frac{dr}{d\theta} \frac{2\dot{r}\dot{\theta}}{r} - r\dot{\theta}^2 \right) \\ &= m \left[ \frac{d^2 r}{d\theta^2} \dot{\theta}^2 - 2 \left( \frac{dr}{d\theta} \right)^2 \frac{\dot{\theta}^2}{r} - r\dot{\theta}^2 \right] = -m\dot{\theta}^2 \left[ \frac{d^2 r}{d\theta^2} - 2 \left( \frac{dr}{d\theta} \right)^2 \frac{1}{r} - r \right] \\ &= m \left( \frac{L}{mr^2} \right)^2 \left[ -2R \cos \theta - 2 \frac{(-2R \sin \theta)^2}{2R \cos \theta} - 2R \cos \theta \right] \\ &= \frac{L^2}{mr^4} \left[ -4R \cos \theta - 4 \frac{R(1 - \cos^2 \theta)}{\cos \theta} \right] = -\frac{L^2}{mr^4} \frac{4R}{\cos \theta} = -\frac{8R^2 L^2}{mr^4} \frac{1}{2R \cos \theta} = -\frac{8R^2 L^2}{mr^5} \end{aligned}$$

4. A bead of mass  $M$  moves without friction along a stationary rod that lies along the horizontal  $x$ -axis. A pendulum of length  $L$  is suspended from this bead with a mass  $m$  at its end. The pendulum swings in the  $xy$ -plane, where  $y$  is the vertical axis. Assuming small oscillations, what is the frequency at which the pendulum oscillates?

(Hint: To write the equation of motion of the pendulum in the noninertial frame of the bead.)



Solution: The equation of motion of the pendulum is

$$\mathbf{e}_\rho : \quad mg \cos \theta - F_T - m\ddot{x} \sin \theta = m(-L\dot{\theta}^2) \quad (1)$$

$$\mathbf{e}_\theta : \quad -mg \sin \theta - m\ddot{x} \cos \theta = mL\ddot{\theta} \quad (2)$$

The equation of motion of the bead is

$$M\ddot{x} = F_T \sin \theta \quad (3)$$

$$(3) - (1) \times \sin \theta - (2) \times \cos \theta :$$

$$(M + m)\ddot{x} = mL\dot{\theta}^2 \sin \theta - mL\ddot{\theta} \cos \theta \quad (4)$$

For small oscillations, using the approximations  $\sin \theta \approx \theta$ ,  $\cos \theta \approx 1$  and neglecting the high order small quantity, equations (2) and (4) can be approximated as

$$-mg\theta - m\ddot{x} = mL\ddot{\theta} \quad (5)$$

and

$$(M + m)\ddot{x} = -mL\ddot{\theta} \quad (6)$$

Substituting  $\ddot{x} = -\frac{m}{M + m}L\ddot{\theta}$  into (5) and after some algebra, we have the small oscillation equation

$$\ddot{\theta} + \frac{M + m}{M} \frac{g}{L} \theta = 0$$

Thus  $\omega = \sqrt{\frac{M + m}{M}} \sqrt{\frac{g}{L}}.$